Sheet 7

1. $y_t = \lambda + y_{t-1} + v_t$; $z_t = \mu + z_{t-1} + w_t$
$u_t \overset{iid}{\sim} (0, \sigma_u^2)$ for $u \in \{v, w\}$; $y_0 = z_0 = 0$; $\{v_t\} \perp\!\!\!\perp \{w_t\}$

a). Note that $y_t = \lambda + (\lambda + y_{t-2} + v_{t-1}) + v_t$, and we can roll back recursively to show that

$$y_t = y_0 + \lambda t + \sum_{i=1}^{t} v_i = \lambda t + \sum v_i$$

So $\mathbb{E}[y_t] = \lambda t + \mathbb{E}[\sum v_i] = \lambda t + \sum \mathbb{E}[v_i]$ by linearity
$= \lambda t$ since each v_i mean zero.

$\text{var}(y_t) = \text{var}(\sum v_i)$ since λt non-random
$= \sum \text{var}(v_i)$ as $v_i \perp\!\!\!\perp v_j$ $i \neq j$
$= t\sigma_v^2$ as ident. dist.

1a. yes, correct!

$\text{cov}(y_s, y_t) = \text{cov}(\sum_{i=1}^{s} v_s, \sum_{i=1}^{t} v_t)$ since λt non-random
$= \sum_{i=1}^{s} \sum_{j=1}^{t} \text{cov}(v_i, v_j)$ but $\text{cov}(v_i, v_j) = 0$ unless $i = j$.
$= \sum_{i=1}^{s} \text{var}(v_s)$ since $s < t$
$= s\sigma^2$

b). $y_t \sim I(1) \iff \Delta y_t \sim I(0)$ i.e. is stationary.

$\Delta y_t := y_t - y_{t-1} = \lambda + (\beta - 1) y_{t-1} + v_t$, where $\delta := \beta - 1$ being zero corresponds to y_t having a unit root.

So, we should run the auxiliary regression $\Delta y_t = \lambda + \delta y_{t-1} + v_t$, and calculate the t-statistic associated with the null of a unit root $H_0: \delta = 0$, i.e. $t = \frac{\hat{\delta}}{se(\hat{\delta})} \sim DF_{trend}$; vs $H_1: \delta < 0$.
($se(\hat{\delta})$ can be found as usual with the sample residual variance.)

We must use tabulated values for c_α since the DF distribution is non-standard. To verify that y_t is $I(1)$ rather than

1b. ok, great answer!

$I(d)$ for some $d > 1$ but possessing a unit root, we should then test for whether Δy_t itself has a unit root, where we'd expect to conclude that it doesn't and is stationary.

Alternatively, we might want to use the KPSS test which reverses the hypothesis such that the null is of stationarity, and the alternative $H_1: y_t \sim I(1)$. In that case, we'd expect to reject the null, but fail to reject it for Δy_t.

With autocorrelated errors, we could either:

- Use the KPSS test, since it estimates HAC variances for v_t
- Use an augmented Dickey-Fuller test by adding lags of y_t to the RHS of the auxiliary regression, in order to capture the temporal dependence so the new errors ε_t are independent
  - ↳ Use AIC/BIC or lag order testing to determine up to what y_{t-p} to include.

c) This is a spurious regression. It arises because even unrelated $I(1)$ process will - due to the nature of their random wandering - have long periods of rising or falling at the same time, which happens regularly enough that a statistically significant relationship emerges.

1c. ok, good!

Adding more observations would not help; we still have this systematic tendency to observe statistically significant relationships even between unrelated processes, in the presence of stochastic trends. Indeed, as $T \to \infty$:

- $|t_0| \to \infty$ rather than being $N(0,1)$
- R^2 follows a non-degenerate limiting distribution despite unrelatedness
- The regression coefficient $\hat{\theta}$ is not consistent for zero but remains a random variable

d) To test for the absence of cointegration, we want to confirm that there does not exist a cointegrating coefficient θ such that

* How does cointegration testing e.g. to establish whether I(2) series are cointegrated and their difference is I(1), etc? *

$x_t := y_t - \theta z_t \sim I(0)$. Assuming we've already established that y_t and $z_t \sim I(1)$, we would then use the E-G two-step procedure:

1. Perform a static regression of y_t onto z_t to obtain an estimate of $\hat{\theta}$ in the model $y_t = \beta_0 + \theta z_t + x_t$
2. Compute the residuals $\hat{x}_t := y_t - \hat{\theta} z_t - \hat{\beta}_0$, and then test for a ~~stationarity~~ a unit root using a DF test with the auxiliary regression $\Delta \hat{x}_t = \delta x_{t-1} + e_t$.

1d. ok, good answer!

3. If we fail to reject the null of a unit root, then conclude that y_t and z_t are not cointegrated, since the residual $\hat{x}_t$ of their weighted difference inherited a stochastic trend.
   ↳ Note that we must use adjusted Engle-Granger cα because the residuals are only estimates of the errors $y_t - \theta z_t$.

2. $$y_t = \underset{-0.2}{\gamma_1} y_{t-1} + \underset{0.4}{\gamma_2} y_{t-2} + \underset{0.3}{\beta_0} x_t + \underset{0.8}{\beta_1} x_{t-1} + \underset{0.6}{\beta_2} x_{t-2} + u_t$$

$u_t \overset{iid}{\sim} N(0, \sigma^2)$, $\{u_t\} \perp\!\!\!\perp \{x_t\}$

a) The impact multiplier $\partial y_t / \partial x_t = \beta_0 = 0.3$.

The total multiplier $m_T := \sum_{j=0}^{\infty} m_j$. Since $m_j := \frac{\partial y_t}{\partial x_{t-j}} = \delta_j$,

from the model's DL form $y_t = \alpha + \sum_{j=0}^{\infty} \delta_j x_{t-j} + u_t$,

$m_T = \sum_{j=0}^{\infty} \delta_j = D(1) = \frac{B(1)}{C(1)}$, where $B(1) = \beta_0 + \beta_1 + \beta_2 = 1.7$ and $C(1) = 1 - \gamma_1 - \gamma_2 = 0.8$

$= 2.125$

The mean lag $:= \sum_{j=0}^{\infty} \frac{m_j}{m_T} j = \frac{D'(1)}{D(1)} = \frac{B'(1)}{B(1)} - \frac{C'(1)}{C(1)}$

$B'(L) = \beta_1 + 2\beta_2$; $C'(L) = -\gamma_1 - 2\gamma_2$, so mean lag $= \frac{2}{1.7} - \frac{-0.6}{0.8}$
$= 1.926$

~~*relationship between β, B and ε~~ ~~defined or derived?~~ ~~[illegible]~~

The median lag $:= \min_{q} \left\{ \frac{\sum_{j=0}^{q} \delta_j}{\sum_{j=0}^{\infty} \delta_j} \geq 0.5 \right\}$.

To find this, we need to write out the full DL form, where
$$y_t = \frac{B(L)}{C(L)} x_t + \frac{\varepsilon_t}{C(L)}, \quad \text{with } D(L) := \frac{B(L)}{C(L)} = \sum_{j=0}^{\infty} \delta_j L^j$$

So $(1-\gamma_1 L - \gamma_2 L^2)(\delta_0 + \delta_1 L + \delta_2 L^2 + \dots) = (\beta_0 - \beta_1 L - \beta_2 L^2)$.
as $C(L) \sum \delta_j L^j = B(L)$.

$$LHS = \delta_0 + (\delta_1 - \delta_0 \gamma_1) L + (-\delta_1 \gamma_1 - \gamma_2 \delta_0 + \delta_2) L^2 + \dots$$

2a. yes, correct!

Matching coefficients, $\delta_0 = \beta_0 = 0.3$
$$\delta_1 - \delta_0 \gamma_1 = \beta_1 \Rightarrow \delta_1 = 0.3 \times -0.2 + 0.8 = 0.74$$
$$\delta_2 - \delta_1 \gamma_1 - \delta_0 \gamma_2 = \beta_2 \Rightarrow \delta_2 = 0.6 + -0.2 \times 0.74 + 0.4 \times 0.3 = 0.572$$
$$\delta_3 - \delta_2 \gamma_1 - \delta_1 \gamma_2 = 0$$
$$\delta_4 - \delta_2 \gamma_2 = 0$$
$$\delta_5 = \delta_6 = \dots = 0$$

Recall that $m_T = 2.125$. So we need the sum to reach 1.06, which happens just after $q = 1$. So the median lag is 2.

def'n of ECM?

~~b) The error correction form of an ARDL(p,q) model is~~
~~$\Delta y_t = \alpha x_t$~~

b) ~~$\Delta y_t := y_t - y_{t-1} = (\gamma_1 - 1) y_{t-1} + \gamma_2 y_{t-2} + \beta_0 x_t + \beta_1 x_{t-1} + \beta_2 x_{t-2} - \beta_0 x_{t-1} + \beta_0 x_{t-1} + u_t$~~

~~$= (\gamma_1 - 1) y_{t-1} + \gamma_2 y_{t-2} + \beta_0 \Delta x_t + (\beta_0 + \beta_1) x_{t-1} + \beta_2 x_{t-2} + u_t$~~

The error correction term is, in general, $y_{t-1} - \theta x_{t-1} - \tau$, since in eq* the long-run relationship is $y = \theta x + \tau$. In this case, $\tau = 0$ since there's no intercept in (2). So the goal is to rewrite this in the form $\Delta y_t = \underbrace{\alpha}_{\text{adj. factor}} (y_{t-1} - \theta x_{t-1}) + SR + u_t$

$$\Delta y_t := y_t - y_{t-1} = (\gamma_1 - 1) y_{t-1} + \gamma_2 y_{t-2} + \beta_0 x_t + \beta_1 x_{t-1} + \beta_2 x_{t-2} + u_t$$
$$+ \gamma_2 y_{t-1} - \gamma_2 y_{t-1} - \beta_0 x_{t-1} + \beta_0 x_{t-1} - \beta_2 x_{t-1} + \beta_2 x_{t-1}$$

$$= (\gamma_1 + \gamma_2 - 1) y_{t-1} - \gamma_2 \Delta y_{t-1} + \beta_0 \Delta x_t + (\beta_0 + \beta_1 + \beta_2) x_{t-1} - \beta_2 \Delta x_{t-1} + u_t$$

2b. yes, correct!

$$= \alpha(y_{t-1} - \theta x_{t-1}) - \gamma_2 \Delta y_{t-1} + \beta_0 \Delta x_{t} - \beta_2 \Delta x_{t-1} + u$$

(*) MT appears!

where $\alpha = \gamma_1 + \gamma_2 - 1 = -0.8$; $\theta = -\dfrac{\beta_0 + \beta_1 + \beta_2}{\gamma_1 + \gamma_2 - 1} = 2.125$ which is exactly MT.

c) $\mathbb{E}[y_t] =: \mu_y$ is the long-run mean in y_t. Since $\{y_t\}$ and $\{x_t\}$ are stationary, we have $\mathbb{E}[\Delta z_t] = 0$ for x and y, at all times t.

(Note that $\Delta z_t := z_t - z_{t-1}$ but for stationary processes $\mathbb{E}[z_t] = \mathbb{E}[z_{t-1}]$.)

So $\mathbb{E}[\Delta y_t] = 0 = \mathbb{E}[\alpha(y_{t-1} - \theta x_{t-1})] - \gamma_2 \mathbb{E}[\Delta y_{t-1}] + \beta_0 \mathbb{E}[\Delta x_{t}] - \beta_2 \mathbb{E}[\Delta x_{t-1}] + \mathbb{E}[u]$

$= \alpha \mathbb{E}[(y_{t-1} - \theta x_{t-1})]$

$\Leftrightarrow \mathbb{E}[y_{t-1}] = \theta \mathbb{E}[x_{t-1}]$ since $\alpha = -0.8 \neq 0$

$= \theta \mu_x$ as stationary

$\therefore \mu_y = \theta \mu_x = 2.125 \mu_x$

2c. yes, correct!

d) We now have $y_t = \underset{-0.2}{\gamma_1} y_{t-1} + \underset{0.3}{\beta_0} x_t + \underset{iid \sim N(0,\sigma_u^2)}{u_t}$ with $x_t = \sum_{i=1}^{t} \underset{iid \sim (0,\sigma_\varepsilon^2)}{\varepsilon_i}$,

(*) how does adding I(1) and I(0) work? You get I(1) I think - but what about I(d) ± I(e) in general? e.g. could we have y_t I(2), then $y_t - \theta x_t \sim I(1)$?

Using the resu[lt ...]

$\Delta y_t = \alpha$ [...]

2d. ok you're on the right track - you need to explicitly show that this linear combination of y_t and x_t is I(0)

where now $\alpha = \gamma_1 - 1 = -1.2$; $\theta = 0.25$. Note that $u_t \sim I(0)$, and $\Delta x_{t-1} = \varepsilon_{t-1} \sim I(0)$.

(*) although what if $y_t \sim I(0)$?

Note that if $y_t \sim I(d)$, then by definition $\Delta y_t \sim I(d-1)$. So for the orders of integration of the RHS and LHS to be equal (which they must), given that the final two RHS terms are I(0), there must be some cancellation of stochastic trend in $y_{t-1} - \theta x_{t-1}$ in order for this to work. So $y_{t-1} - \theta x_{t-1} \sim I(0)$ and thus they are indeed cointegrated.

3. $y_t = \gamma y_{t-1} + \beta x_t + \delta x_{t-1} + \mu + u_t$ $\quad \{u_t\} \perp\!\!\!\perp \{v_t\} \overset{iid}{\sim} N(0, \sigma_u^2)$

$x_t = \alpha x_{t-1} + \eta + v_t$

a) $\Delta y_t := y_t - y_{t-1} = (\gamma - 1) y_{t-1} + \beta x_t + \delta x_{t-1} + \mu + u_t$

$\qquad - \beta x_{t-1} + \beta x_{t-1}$

3a. yes, correct!

$\beta \Delta x_t + (\gamma - 1) y_{t-1} + (\beta + \delta) x_{t-1} + \mu + u_t$

$= \xi \Delta x_t + K(y_{t-1} - \tau x_{t-1} - \nu) + \varepsilon_t$

where $\xi = \beta$, $K = \gamma - 1$, $\tau = \frac{\beta + \delta}{\gamma - 1}$, $\nu = \frac{\mu}{\gamma - 1}$

b) $x_t = 0.6 x_{t-1} + 0.8 + v_t$.

i. $\mathbb{E}[x_t] = \mathbb{E}[x_{t-1}]$ by stationarity

$\therefore \mu_x = 0.6 \mu_x + 0.8 + \mathbb{E}[v_t]$

$\mu_x = 0.8 / 0.4 = 2$

3b. yes, correct!

ii. $Var(x_t) = var(x_{t-1})$ by stationarity.

$\therefore \sigma_x^2 = 0.6^2 \sigma_x^2 + \sigma_v^2$ $\quad$ assuming that $v_t \perp\!\!\!\perp I_{t-1}$

$\sigma_x^2 = 0.8 / 0.64 = 1.25$

iii. $\mathbb{E}[y_t] = \mathbb{E}[y_{t-1}]$ by stationarity.

$\therefore \mu_y = 0.8 \mu_y + 0.8 \mu_x + \cancel{0.4} + \mathbb{E}[u_t]$

$= \cancel{8(0.8 \cdot 2 + 2.4)}$ $5 \cdot (0.8 \times 2 + 0.4) = 10$

c) ~~$y_t = 0.8 y_{t-1} + 0.8 x_{t-1} + u_t$; $x_t = x_{t-1} + v_t$~~

~~$\Delta y_t = 0.2(y_{t-1} + 4x_{t-1})$~~ $+ u_t$.

~~$\therefore \Delta y_t = -0.2(y_{t-1} + 4x_{t-1} + 4v_t) + u_t$~~

c). Let $z_t := y_t - \tau x_t$; $\Delta z_t = \Delta y_t - \tau \Delta x_t$.

From (3), $x_t = x_{t-1} + v_t$; $\Delta x_t = v_t$

From (4) $\Delta y_t = K(y_{t-1} - \tau x_{t-1}) + u_t$ since $\beta = \mu = 0$.

~~$\therefore \Delta z_{t} + \tau \Delta z_t = K z_{t-1} + u_t$.~~

$\therefore \Delta z_t + \tau \Delta x_t = K(y_{t-1} - \tau x_{t-1}) + u_t$. by subs for Δy_t.

$$\Delta z_t = K z_{t-1} \underbrace{- \tau v_t + u_t}_{} \quad \text{by subs for } \Delta x_t$$

$$=: K z_{t-1} + e_t$$

i.e. $z_t = (K+1) z_{t-1} + e_t$. 3c. yes, correct!

In this case, $K + 1 = \gamma = 0.8$ which in absolute value is < 1.
So in the long run, $z_t = y_t - \tau x_t$ is stationary, i.e. $\sim I(0)$.
* Do you need to prove $E[z_t]$ and $Var(z_t)$, etc, here? *

$\Delta x_t =$ not required i think

x_t has a unit root but v_t is stationary, so $x_t \sim I(1)$.
We've already seen that $z_t = y_t - \tau x_t \sim I(0)$. For this to be, τx_t's stochastic trend must cancel out a stochastic trend in y_t. So y_t too must be $\sim I(1)$, and they are indeed cointegrated.

(How would you argue that y_t can't be $I(2)$, since then z_t would be $I(2)$ as you can't cancel higher-order processes with lower-order ones?) not the best with time series, so please discuss this with the tutor!

d). $\therefore$ (4) ~~now states that $\Delta y_t = $~~ is no longer valid, because τ and ν have $\gamma - 1 = 0$ in the denominator and are thus undefined. Also, $K = \gamma - 1 = 0$ so the error correction term simply disappears. When y_t has a unit root, shocks have permanent effects. Since y_t doesn't affect x_t, ~~there's nothing that~~ (only vice versa), there's no mechanism through which x_t and y_t can have a constant long-term relationship $y_t - \tau x_t$, since y_t's ~~is not stationary~~ deviations due to shocks u_t persisting.

correct!

$= \beta x_t + \delta x_{t-1} + \mu + u_t$

Δy_t doesn't depend on eq* deviations, so error correction is not possible.

ii. $\Delta y_t = 0.4x_t - 0.4x_{t-1} + u_t$ yes, correct

$= 0.4\,\Delta x_{t} + u_t$; $\Delta x_{t} = v_t$ as before.

$\because\ e_t$.

So $y_t, x_t \sim I(1)$. But since $y_t = y_{t-1} + 0.4(x_t - x_{t-1}) + u_t$ has its own unit root, there's no linear combination with x_t that would cancel that stochastic trend. e.g. let $z_t := y_t - \theta x_t$, then $\Delta z_t = \Delta y_t - \theta \Delta x_t = 0.4 v_t + u_t - \theta v_t$. We could eliminate x's stochastic trend by letting $\theta = 0.4$, but u_t remains, so ~~the differenced linear combination is stationary not constant and~~ $z_t = z_{t-1} + u_t \sim I(1)$, not $I(0)$.

iii. $\Delta y_t = 0.8x_{t-1} + u_t$

We know $x_t \sim I(1)$. But this means Δy_t is $I(1)$:

$$\Delta^2 y_t = \Delta y_t - \Delta y_{t-1} = 0.8(x_{t-1} - x_{t-2}) + u_t - u_{t-1}$$
$$= 0.8\,\Delta x_{t-1} + u_t - u_{t-1}$$

which is $I(0)$ since it's a sum of stationary variables $v_t, u_t, -u_{t-1}$.

So $y_t \sim I(2)$, while $x_t \sim I(1)$. So they couldn't be cointegrated since they're different $I(d)$.

3d. well done and well answered!